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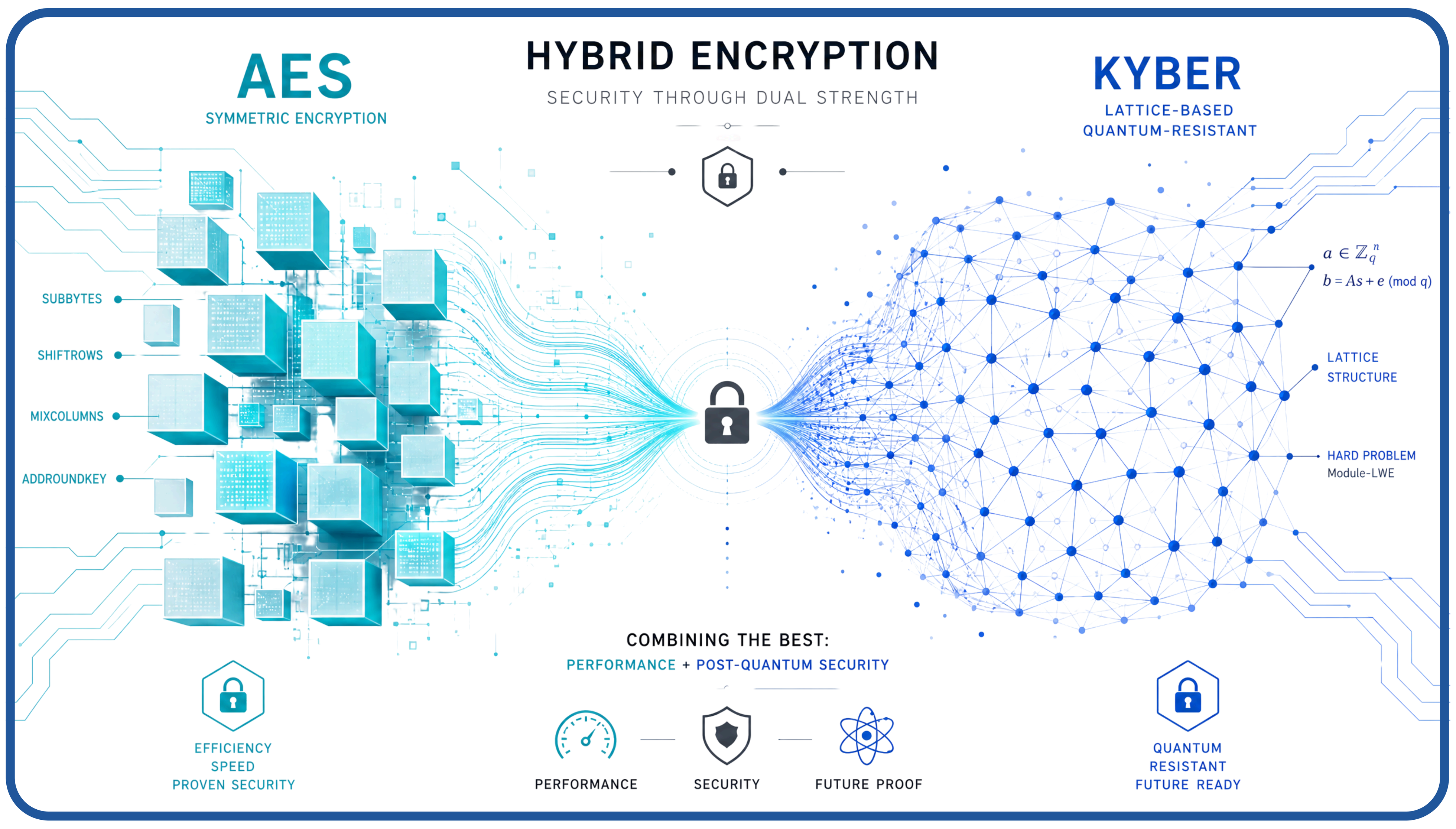
INTRODUCTION

As quantum computing advances, traditional encryption algorithms such as RSA and ECC become increasingly vulnerable, creating an urgent need for quantum-resistant security solutions. SVALINN addresses this challenge through a hybrid cryptographic framework that combines post-quantum algorithms with trusted symmetric encryption to provide secure and future-ready communication against both classical and quantum-based threats.

METHODOLOGY

The proposed system implements a hybrid encryption architecture combining AES-256 symmetric encryption with CRYSTALS-Kyber secure key encapsulation.
Messages and files are encrypted locally on the user's device before transmission. The encrypted data is then securely synchronized through Firebase real-time services without exposing plaintext information to the server.
The platform also utilizes Flutter for cross-platform development, Python for cryptographic operations, and SQLite for secure local storage.



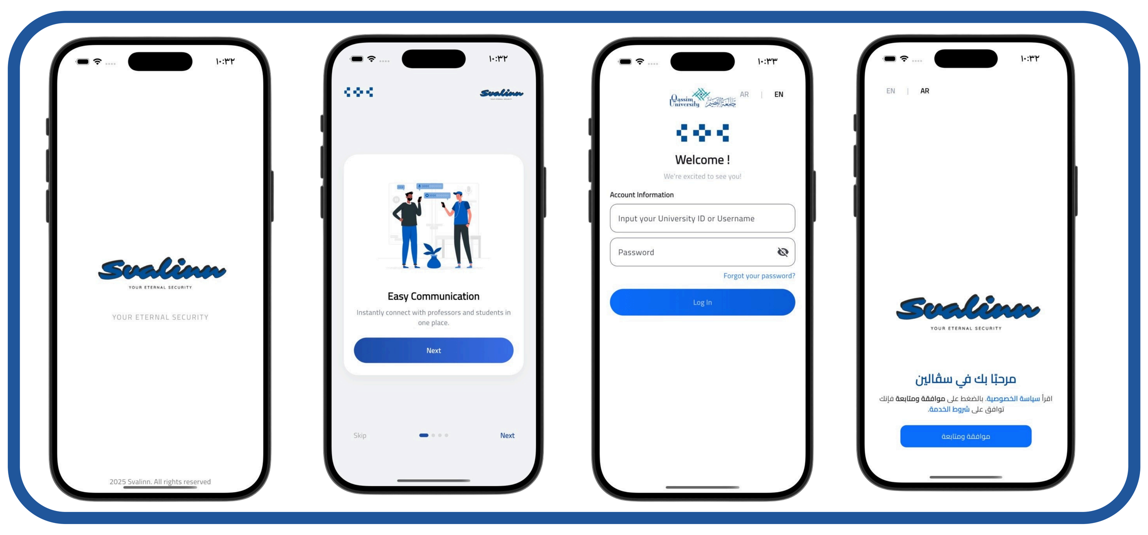
RESULTS

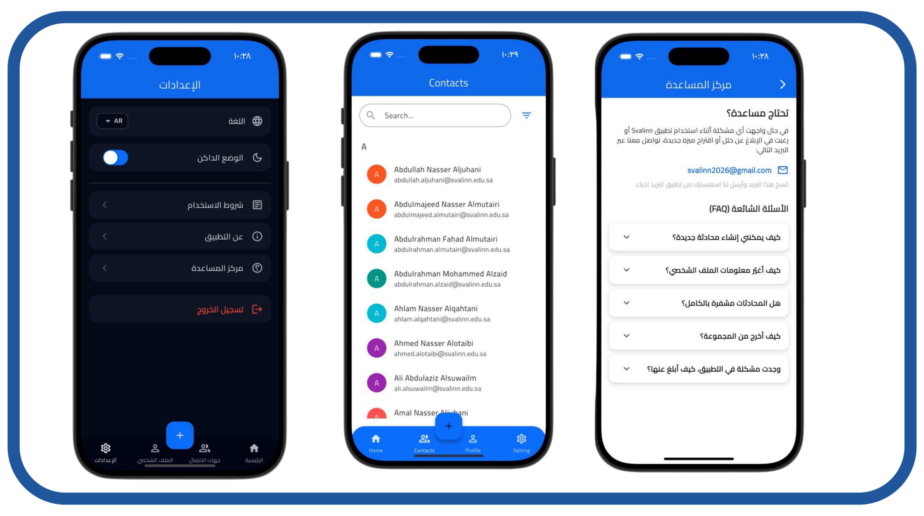
Proof of Concept
Developed a cross-platform application with real-time encrypted communication using quantum-resistant key exchange.

Performance Validation
Achieved efficient key generation and low-latency encryption, proving practical feasibility.
Security Analysis
Demonstrated resistance against classical and simulated quantum attacks.
Framework Success
Post-quantum cryptography can be integrated into modern systems without reducing usability.
Conceptual Innovation
SVALINN emphasizes the need for future-ready cybersecurity solutions.

COMPARISON

TRADITIONAL SYSTEMS	Svalinn
Classical Encryption	Post-Quantum Encryption
Vulnerable to Quantum Threats	Quantum Resistant
Limited Privacy Protection	Full End-to-End Encryption
Centralized Trust Risks	Secure Hybrid Architecture





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